Benchmarking General and Medical Small Language Models for Spanish Obstetric Question Answering: The Impact of Fine-Tuning

Jhon H. Mejía-Muñoz
Facultad de Ingeniería, IUE
Envigado, Colombia
0009-0008-7768-4072

Nicolás Hoyos-Giraldo
Facultad de Ingeniería, IUE
Envigado, Colombia
0009-0008-7768-4072

Daniel Betancur-Vasquez
Facultad de Ingeniería, IUE
Envigado, Colombia
0000-0003-4834-7268

Kevin D. Trujillo
Facultad de Ingeniería, IUE
Envigado, Colombia
0009-0001-7678-2681

***Abstract*—Small Language Models (SLMs) offer a computationally efficient alternative to large language models for clinical question answering; however, evidence on domain adaptation for Spanish obstetric applications remains limited. This study compares the general-purpose Gemma 4 model and the medically pretrained MedGemma 1.5 before and after grounded QLoRA adaptation using MaternaQA-es, while examining No RAG, Hybrid retrieval, and independent HyDE as experimental factors. All 24 scenario-condition runs were evaluated across two 10-question scenarios, yielding 240 final condition-item outputs. Within this exploratory evaluation, on MaternaQA-es, Gemma 4 Base with Hybrid retrieval achieved the highest observed Answer Correctness (0.649), while Gemma 4 QLoRA with No RAG and Hybrid retrieval achieved the highest Answer Relevancy (0.776) and Semantic Similarity (0.838), respectively. On the obstetric challenge set, Gemma 4 Base with Hybrid retrieval achieved the highest Answer Correctness (0.457), MedGemma 1.5 QLoRA with No RAG the highest Answer Relevancy (0.913), and Gemma 4 Base and Gemma 4 QLoRA with HyDE tied in Semantic Similarity (0.707). HyDE achieved higher RAGAS Context Precision than Hybrid in seven of eight model-scenario comparisons; effects on answer quality were heterogeneous and model-dependent. Under No RAG, QLoRA increased Answer Relevancy in all four model-scenario comparisons, while changes in Answer Correctness and Semantic Similarity were mixed. Given the limited evaluation of 10 questions per scenario, these findings should be interpreted as preliminary descriptive patterns.**

***Index Terms*—Small language models, QLoRA, retrieval-augmented generation, HyDE, obstetric question answering, Spanish clinical NLP, model benchmarking.**

I. Introduction

The rapid adoption of Large Language Models (LLMs) has transformed the development of clinical decision-support systems and medical question-answering (QA) applications. Nevertheless, their computational requirements limit deployment in many healthcare environments, motivating increasing interest in Small Language Models (SLMs), which offer lower inference costs, reduced hardware requirements, and easier local deployment while maintaining competitive performance in domain-specific tasks [1], [2]. In medical applications, supervised fine-tuning has emerged as a practical strategy for adapting compact language models to specialized clinical domains while preserving computational efficiency.

Recent studies have shown that domain-specific fine-tuning and retrieval can improve medical question answering under resource constraints [3]. Work on fine-tuned clinical embedding models likewise demonstrates the value of domain-specific retrieval, but remains outside Spanish obstetric QA [4]. Recent obstetric applications include simulated-case decision support, emergency-response evaluation, and neonatal growth prediction [5]–[7]. Reviews of maternal-health RAG emphasize the lack of standardized benchmarks and clinically validated evaluation protocols [8]. This gap is especially relevant in obstetric care, where inaccurate recommendations may affect maternal and fetal safety [9].

An important question therefore remains: does medical pretraining continue to provide measurable advantages after domain-specific fine-tuning? Medically pretrained SLMs should encode biomedical terminology, clinical concepts, and reasoning patterns. Yet, after both a general-purpose and a medical model receive identical specialized supervision, it remains unclear whether prior medical knowledge provides an additional benefit or whether the general-purpose model can achieve comparable or superior performance.

To investigate this question, this study uses MaternaQA-es, a Spanish obstetric QA resource containing 5,727 synthetic question-answer pairs generated from 63 Spanish-language clinical documents. The dataset is publicly available through Hugging Face [10]. The benchmark compares Gemma 4 and MedGemma 1.5 before and after identical grounded QLoRA adaptation.

This work makes three contributions. First, it compares general-purpose and medically pretrained SLMs for Spanish obstetric QA. Second, it evaluates the effect of domain-specific QLoRA adaptation under matched training conditions. Third, it provides a reproducible experimental design that incorporates retrieval-augmented inference and item-level evaluation while explicitly separating completed runs from planned factorial conditions.

II. Methodology

A. Study Design and Research Questions

The study uses a controlled, offline factorial benchmark. Model family has two levels: Google Gemma 4 E2B instruction-tuned (`google/gemma-4-E2B-it`) and Google MedGemma 1.5 4B instruction-tuned (`google/medgemma-1.5-4b-it`). Parameter adaptation has two levels: base weights without an adapter and grounded QLoRA. Retrieval strategy has three planned levels: the No RAG control, canonical Hybrid retrieval, and canonical independent HyDE. Their Cartesian product defines 12 planned conditions. Throughout the paper, "Base" means no QLoRA adapter, and "No RAG" means no retrieved context (the model answers from the question alone); the two are independent and can be combined.

The unit of analysis is an item-level paired `qa_id`. The primary QLoRA estimand is the paired difference between QLoRA and base outputs within a fixed model family and retrieval strategy. Retrieval contrasts compare strategies within a fixed family and adaptation state. The interaction estimand is the change in the QLoRA effect across retrieval strategies:

$$\Delta = (Y_{Q,H} - Y_{B,H}) - (Y_{Q,N} - Y_{B,N}). \tag{1}$$

Here Q denotes QLoRA, B denotes the base model, H denotes Hybrid retrieval, and N denotes the No RAG condition. Analogous contrasts are defined for HyDE and for the two model families.

B. Dataset, Provenance, and Split Integrity

MaternaQA-es contains 5,727 Spanish obstetric question-answer pairs derived from 63 clinical documents. Its document-level partitions contain 52 training PDFs (5,093 items), 2 validation PDFs (306 items), and 3 held-out test PDFs (328 items). The exploratory evaluation subset comprised the first 10 records of the held-out test file used in the experiment, rather than the complete test partition. Test sources do not overlap training or validation sources, and `qa_id` is retained as the pairing key [10].

The primary grounded adaptation uses the `sft_grounded` representation, in which the source context associated with an item is concatenated with the question. The term base indicates whether an adapter is loaded; it does not indicate that the input is closed-book when context is otherwise present.

The same 10 held-out test questions were processed across all model and retrieval conditions. The 2,268-chunk retrieval corpus includes the source material associated with these questions. Therefore, the RAG evaluation measures the system's ability to retrieve and use relevant evidence when that evidence is available in the corpus, rather than strict generalization to completely unseen source documents. This inference-time access is distinct from leakage into model parameters, because the corresponding test documents were not used during fine-tuning.

C. Protocol Evaluation Scenarios

Two protocol scenarios define the evaluation inputs. The first is a 10-item obstetric challenge set represented by `datasets/sample10.jsonl`; these records do not provide document-level reference chunk identifiers. The second consists of 10 processed questions from the held-out MaternaQA-es test partition, with QA metadata and source-reference information. MaternaQA-es is publicly available on Hugging Face [10].

The challenge set supports answer-quality evaluation but not gold document-level retrieval recall. The processed MaternaQA-es subset supports both response metrics and reference-chunk checks. Each scenario used 10 questions across all 12 conditions. MaternaQA-es records retain source-reference metadata, but the test prompt contains only the question before retrieval. The `sample10` records likewise contain the question but do not provide gold reference-chunk identifiers. Hybrid and HyDE add retrieved corpus contexts, whereas No RAG receives none. These differences affect which retrieval checks are available for each scenario.

Therefore, the two scenarios are treated as complementary evaluation settings rather than directly comparable samples from the same input distribution. Absolute metric differences between MaternaQA-es and the obstetric challenge set are not interpreted as effects of fine-tuning or retrieval, because scenario membership is confounded with differences in input construction and source-context availability. Model and retrieval comparisons are consequently emphasized within each scenario.

D. Model Families and Experimental Conditions

The benchmark compares Google Gemma 4 E2B instruction-tuned with Google MedGemma 1.5 4B instruction-tuned. Base conditions load the published model without an adapter. QLoRA conditions load `iue-edu/MaternaCare-ES-gemma4-qlora` or `iue-edu/MaternaCare-ES-medgemma-qlora` on the corresponding base model. The answer model, adaptation state, and test items are held fixed across retrieval-strategy comparisons within each model family; the retrieval strategy determines whether retrieved evidence is supplied.

Table I lists the complete prespecified design. The final result artifacts covered No RAG, Hybrid, and HyDE for all four model/adaptation combinations in both scenarios; therefore, all 24 scenario-condition runs were represented in the analysis.

E. Grounded QLoRA Fine-Tuning

QLoRA [11] is the parameter-adaptation method in the benchmark: trainable low-rank adapters on quantized base weights, which keeps the base model as a separately identifiable initialization. Fig. 1 contrasts it with full fine-tuning, and Table II summarizes the configuration used for both adapters, including the shared quantization and BF16 computation settings.

Both model families used the same MaternaQA-es training and validation partitions and the same grounded input representation. To maintain a controlled comparison between model

TABLE I
PRESPECIFIED FACTORIAL EXPERIMENT MATRIX

ID	Model family	Adaptation	Retrieval
C01	Gemma 4	Base	No RAG
C02	Gemma 4	Base	Hybrid
C03	Gemma 4	Base	HyDE
C04	Gemma 4	QLoRA	No RAG
C05	Gemma 4	QLoRA	Hybrid
C06	Gemma 4	QLoRA	HyDE
C07	MedGemma 1.5	Base	No RAG
C08	MedGemma 1.5	Base	Hybrid
C09	MedGemma 1.5	Base	HyDE
C10	MedGemma 1.5	QLoRA	No RAG
C11	MedGemma 1.5	QLoRA	Hybrid
C12	MedGemma 1.5	QLoRA	HyDE

Note. The complete design is 2 model families × 2 adaptation states × 3 retrieval strategies; all 12 conditions were observed in each of the two evaluation scenarios. *Base* loads the published weights without an adapter, and *QLoRA* loads the grounded adapter described in Section II-E. *No RAG* supplies no retrieved context, whereas *Hybrid* and *HyDE* are the two canonical retrieval strategies of Section II-F.

TABLE II
GROUNDED QLORA CONFIGURATION USED FOR BOTH MODEL FAMILIES

Parameter	Shared setting
Base model	Gemma 4 E2B: `google/gemma-4-E2B-it` MedGemma 1.5 4B: `google/medgemma-1.5-4b-it`
Training representation	sft_grounded
Train / validation examples	5,093 / 306
Quantization	4-bit NF4, double quantization
LoRA rank (r)	16
LoRA alpha	16
LoRA dropout	0.05
Target modules	All linear (all-linear)
Learning rate	2×10^{-4}
Batch size	1 per device
Gradient accumulation steps	8
Effective batch size	8
Epochs / training steps	2 / 1,274
Optimizer	AdamW 8-bit
LR scheduler	Cosine
Maximum sequence length	1,024
Packing	Disabled
Random seed	3407

Note. Shared settings apply to Gemma 4 E2B and MedGemma 1.5 4B; only the base model differs, as identified above.

initializations, the QLoRA hyperparameters and optimization settings were held constant across both models.

The `sft_grounded` examples use the dataset's conversational `Contextofuente` representation, in which source context is presented with the question. This differs from retrieval at inference: training context is pre-associated with each example, whereas RAG selects contexts from the 2,268-chunk corpus for each question. The adapter is loaded only in QLoRA conditions.

F. Retrieval-Augmented Generation Protocol

RAG is the second core method [12]. The protocol treats retrieval strategy as a reproducible experimental factor. No RAG, Hybrid retrieval, and independent HyDE differ in how evidence is prepared before the answer model receives the question and context. Hypothetical text is used only for query expansion.

1) Corpus Provenance and Chunk Construction: The separate retrieval corpus contains 2,268 chunks from curated obstetric source documents and is used for dense, sparse, and fused retrieval. The `clinical_score` is a corpus-filtering signal.

$$
\begin{aligned}
S(x) &= \max\{0,\, L(x) + A(x) + B(W) - P(R)\}, \qquad (2)\\
L(x) &= \sum_{t \in T,\ t \subseteq \mathrm{norm}(x)} \big[1 + \mathbf{1}(t \text{ has a space})\big],\\
A(x) &= \sum_{a \in A,\ a \subseteq \mathrm{norm}(x)} 1,\\
B(W) &= 2\mathbf{1}(W \geq 120) + 2\mathbf{1}(W \geq 350),\ P(R) = 3\mathbf{1}(R \geq 0.35).
\end{aligned}
$$

Here $\mathrm{norm}(x)$ applies Unicode normalization, removes combining marks, lowercases text, replaces non-ASCII letters and punctuation with spaces, and collapses whitespace. T is the clinical-term set, A the action-term set, W the Unicode word count, and R the ratio of reference-matching nonempty lines to all nonempty lines.

2) Dense and Sparse Retrieval: Dense retrieval uses `BAAI/bge-m3` [13]. SentenceTransformer produces normalized dense vectors on CPU with `batch_size=32`. The persistent local vector table is LanceDB [14]. Because no approximate vector index is created, search is exhaustive exact cosine search. The default dense retrieval depth is $k = 5$.

Sparse retrieval is Okapi BM25 (`rank_bm25/BM25Okapi`) over Unicode lowercase word tokens. Retrieval outputs are recorded before answer generation so retrieval quality and latency can be analyzed separately from response generation.

3) Canonical Hybrid Retrieval: Hybrid retrieval fuses dense and BM25 rankings by reciprocal rank fusion with `rank_constant=60` [15]. For a requested final depth k, the candidate depth is:

$$k_c = \min(|C|,\, \max(4k, 50)). \qquad (3)$$

Ties are broken by stable chunk ID, and the final top-k chunks are supplied to the answer model.

4) Canonical Independent HyDE Retrieval: HyDE [16] uses a dedicated hypothetical-document generator, separate from the answer model, to produce a Spanish clinical-document passage from the question. The retained runs used the OpenAI provider with `gpt-5-mini-2025-08-07`. Hypothetical documents were generated once per question and cached for reuse across answer-model conditions; dense retrieval then embeds that text with BGE-M3 and searches the corpus by exact cosine similarity. Only the retrieved corpus chunks, not the hypothetical text, are supplied as evidence to the answer model [17]. The cache is separated by dataset mode, provider, and generator model, and row reuse checks `qa_id` and a question hash.

5) No RAG Control: The No RAG control returns no retrieved contexts and omits the retrieved-context block from

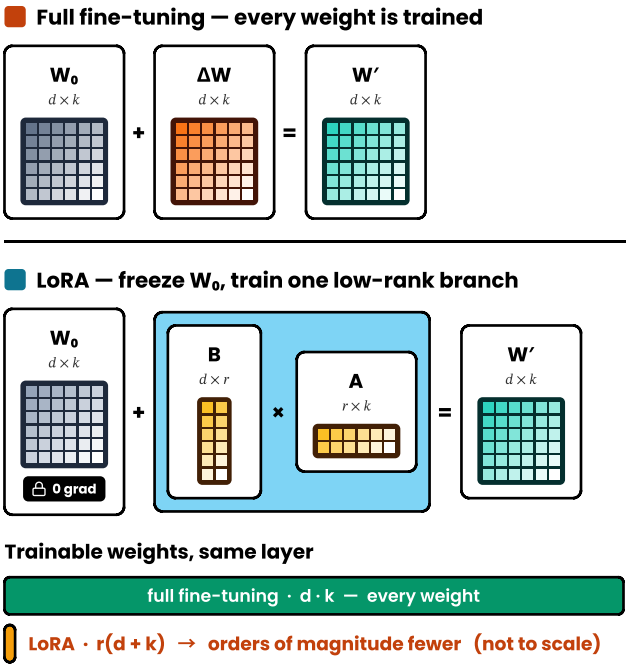


Fig. 1. Comparison of full fine-tuning and LoRA adaptation.

the answer prompt. Context-dependent retrieval metrics are therefore not defined and are not assigned a value of zero.

G. Inference and Prompt Controls

All answer prompts are written in Spanish. RAG prompts contain a clearly delimited context block with positional labels and the question. Provenance is retained in benchmark records rather than asserted as content in the answer prompt. The No RAG condition receives the same question formatting and chat template but omits the context block.

Inference uses the model's chat template and greedy decoding by default, with a maximum of 2,048 newly generated tokens. Sampling is disabled unless a run explicitly records a different configuration. QLoRA conditions load the declared adapter on the corresponding base model; base conditions do not. Retrieval generation and answer generation remain separate stages.

H. Evaluation Framework and Metrics

RAGAS 0.4.3 [18] is used as an external automatic evaluation layer. Response-quality, retrieval-quality, and operational measurement domains are kept separate before paired contrasts and interaction analysis.

1) Retained Fine-Tuning-Output Evaluator: The retained evaluator uses RAGAS 0.4.3 with `gpt-5.4-mini` as evaluator, `text-embedding-3-large` for embedding-based evaluation, and a maximum completion length of 2,048 tokens. The retained metrics are Faithfulness, Answer Relevancy, Answer Correctness, and Semantic Similarity.

2) RAG Benchmark Evaluator: The RAG evaluator uses six metrics: Context Precision, Context Recall, Faithfulness, Answer Relevancy, Answer Correctness, and Semantic Similarity, as defined in RAGAS.

Operational variables include input, output, and total token counts; retrieval, generation, and end-to-end latency; and output tokens per second. Wall-clock output rate is descriptive and not treated as hardware-independent throughput.

$$\mathrm{TPS} = \frac{N_{\mathrm{out}}}{T_{\mathrm{gen}}}. \tag{4}$$

I. Analysis Plan

The primary analysis is descriptive and item-level. Means are computed from the 10 question-level scores for each applicable metric within each balanced scenario; context-dependent metrics are not applicable to No RAG. Comparisons are exploratory. QLoRA is compared with base within each retrieval strategy and family; retrieval strategies are contrasted within adaptation and family; and model families are compared under matched conditions.

The summary used the same 10 `qa_id` values shared by all 12 conditions in each scenario.

Approximate marginal 95% Student-t confidence intervals were calculated as $\bar{x} \pm t_{0.975,n-1}\, s/\sqrt{n}$ from exported means, sample standard deviations, and counts ($n = 10$). This independent-question approximation is limited by nonrandom selection and shared source documents. Intervals for all metrics and unstandardized mean differences accompany the reproducibility files. Paired tests, bootstrap intervals, and standardized paired effects require current question-level scores, unavailable in the aggregate exports, and are not reported.

J. Reproducibility and Experimental Scope

Reproducibility requires recording the MaternaQA-es revision, retrieval-corpus hash, complete configuration, prompts, model and adapter identifiers, HyDE provider settings, cache identity, evaluator version, and decoding settings. The pinned BGE-M3 revision and deterministic RRF tie-breaking are part of this record. Reporting follows the transparency goals of TRIPOD-LLM and Data Statements for NLP [19], [20].

QLoRA fine-tuning and local neural-model computations were performed on an NVIDIA GeForce RTX 5070 Ti Laptop GPU with 12 GB of dedicated VRAM. The comparison comprised 20 questions—10 from MaternaQA-es and 10 from `sample10`—evaluated across all 12 conditions, and all principal comparisons are made within each scenario.

III. Results

A. Available Runs and Analysis Set

The final analysis comprised 240 condition-item outputs: 120 for the MaternaQA-es subset and 120 for the obstetric challenge set. Each scenario contained 10 questions evaluated across all 12 conditions, yielding 10 final responses per condition. All automatic metrics applicable to each condition were available; no scores were imputed.

TABLE III
MEAN AUTOMATIC METRICS FOR THE 10 SHARED QUESTIONS IN EACH SCENARIO

Model	State	Strategy	Correct.	Relev.	Sem. sim.	Faith.	C. prec.	C. recall
Panel A. MaternaQA-es test subset								
Gemma 4	Base	No RAG	0.372	0.696	0.704	—	—	—
Gemma 4	Base	Hybrid	**0.649**	0.669	0.820	**0.967**	0.908	0.800
Gemma 4	Base	HyDE	0.560	0.685	0.796	0.954	**0.950**	0.850
Gemma 4	QLoRA	No RAG	0.384	**0.776**	0.726	—	—	—
Gemma 4	QLoRA	Hybrid	0.605	0.759	**0.838**	0.883	0.908	0.800
Gemma 4	QLoRA	HyDE	0.637	0.639	0.818	0.882	**0.950**	0.750
MedGemma 1.5	Base	No RAG	0.388	0.611	0.714	—	—	—
MedGemma 1.5	Base	Hybrid	0.599	0.648	0.794	0.897	**0.950**	**0.900**
MedGemma 1.5	Base	HyDE	0.595	0.737	0.807	0.842	**0.950**	**0.900**
MedGemma 1.5	QLoRA	No RAG	0.330	0.683	0.720	—	—	—
MedGemma 1.5	QLoRA	Hybrid	0.609	0.700	0.826	0.883	0.908	0.850
MedGemma 1.5	QLoRA	HyDE	0.574	0.674	0.828	0.859	**0.950**	**0.900**
Panel B. Obstetric challenge set								
Gemma 4	Base	No RAG	0.359	0.716	0.651	—	—	—
Gemma 4	Base	Hybrid	**0.457**	0.554	0.689	0.966	0.800	0.400
Gemma 4	Base	HyDE	0.335	0.658	**0.707**	0.902	0.958	0.200
Gemma 4	QLoRA	No RAG	0.205	0.845	0.634	—	—	—
Gemma 4	QLoRA	Hybrid	0.349	0.736	0.693	0.838	0.775	0.400
Gemma 4	QLoRA	HyDE	0.399	0.672	**0.707**	0.919	0.933	0.200
MedGemma 1.5	Base	No RAG	0.287	0.832	0.631	—	—	—
MedGemma 1.5	Base	Hybrid	0.402	0.368	0.622	0.850	0.842	**0.500**
MedGemma 1.5	Base	HyDE	0.260	0.545	0.639	0.850	**1.000**	0.200
MedGemma 1.5	QLoRA	No RAG	0.234	**0.913**	0.649	—	—	—
MedGemma 1.5	QLoRA	Hybrid	0.455	0.787	0.678	0.732	0.842	0.400
MedGemma 1.5	QLoRA	HyDE	0.316	0.725	0.668	0.771	**1.000**	0.200

Note. Correct. = answer correctness; Relev. = answer relevancy; Sem. sim. = semantic similarity; Faith. = faithfulness; C. prec. = context precision; C. recall = context recall. Dashes mark non-applicability for No RAG. All conditions yielded 10 responses; $n = 10$ for every applicable metric; context metrics are not applicable to No RAG. Bold values are panel maxima.

B. MaternaQA-es Subset

Gemma 4 Base with Hybrid retrieval achieved the highest observed mean Answer Correctness (0.649), whereas Gemma 4 QLoRA with No RAG achieved the highest Answer Relevancy (0.776), and Gemma 4 QLoRA with Hybrid retrieval achieved the highest Semantic Similarity (0.838). Under No RAG, QLoRA changed Gemma 4 Answer Correctness, Answer Relevancy, and Semantic Similarity by +0.012, +0.080, and +0.022, respectively. For MedGemma 1.5, the corresponding changes were −0.058, +0.072, and +0.006.

Relative to No RAG, Hybrid and HyDE increased Answer Correctness and Semantic Similarity in all four model-state combinations. Answer Relevancy increased in two of four combinations with Hybrid and one of four with HyDE. Compared directly with Hybrid, HyDE increased Answer Correctness in one of four combinations, Answer Relevancy in two, and Semantic Similarity in two, indicating heterogeneous effects across models and metrics.

C. Obstetric Challenge Set

Within the evaluated obstetric challenge subset, Gemma 4 Base with Hybrid retrieval achieved the highest observed mean Answer Correctness (0.457), and MedGemma 1.5 QLoRA with No RAG achieved the highest Answer Relevancy (0.913). Gemma 4 Base and Gemma 4 QLoRA with HyDE tied for the highest Semantic Similarity (0.707). Under No RAG, QLoRA changed Answer Correctness, Answer Relevancy, and Semantic Similarity by −0.154, +0.129, and −0.017 for Gemma 4, and by −0.054, +0.081, and +0.017 for MedGemma 1.5.

Relative to No RAG, Hybrid increased Answer Correctness in all four model conditions but reduced Answer Relevancy in every condition. Semantic Similarity increased for both Gemma 4 conditions and MedGemma 1.5 QLoRA, but decreased for MedGemma 1.5 Base.

Relative to No RAG, HyDE increased Answer Correctness for Gemma 4 QLoRA (+0.194) and MedGemma 1.5 QLoRA (+0.082), but reduced it for Gemma 4 Base (−0.024) and MedGemma 1.5 Base (−0.028). Answer Relevancy decreased in all four conditions, while Semantic Similarity increased in all four.

Compared directly with Hybrid retrieval, HyDE improved Answer Correctness for Gemma 4 QLoRA (+0.050) and Answer Relevancy for Gemma 4 Base (+0.104) and MedGemma 1.5 Base (+0.177). Semantic Similarity increased in three of four model conditions. These descriptive patterns show that higher retrieval scores did not guarantee uniformly better response metrics.

D. Retrieval-Grounding Metrics

In the MaternaQA-es subset, Hybrid Context Precision ranged from 0.908 to 0.950, Context Recall from 0.800 to 0.900, and Faithfulness from 0.883 to 0.967. HyDE Context Precision was 0.950 in all four model states, Context Recall ranged from 0.750 to 0.900, and Faithfulness from 0.842 to 0.954. HyDE exceeded the corresponding Hybrid Context Precision in three of four model states and increased Context Recall in two of four.

In the obstetric challenge set, Hybrid Context Precision ranged from 0.775 to 0.842, Context Recall from 0.400 to

0.500, and Faithfulness from 0.732 to 0.966. HyDE Context Precision ranged from 0.933 to 1.000, Context Recall was 0.200 in all four model states, and Faithfulness ranged from 0.771 to 0.919. HyDE achieved higher Context Precision than Hybrid for all four model states, while Context Recall was lower for all four.

Across both scenarios, the 20 unique uncached HyDE generations averaged 18.5 seconds, 67.1 input tokens, and 1,437.3 API output tokens. The same hypothetical documents were then reused from cache across the remaining answer-model conditions, reducing model-dependent variation in the query-expansion stage.

E. Uncertainty of Answer Correctness

Table IV reports broad approximate marginal 95% confidence intervals for Answer Correctness, indicating substantial item-level variability. They do not quantify paired differences, and overlap is not a significance test. Rankings remain exploratory; statistically significant superiority is not established.

F. Illustrative Grounded-Answer Audit

Three archived base–QLoRA pairs from `master_eval.csv` (`sft_grounded`, supplied context) were purposively audited, separately from Table III. Table V gives English summaries; Spanish records and identifiers accompany the reproducibility files. Source-support checks are not clinician adjudication, and scores are not pooled.

IV. Discussion

Across the evaluated 20-question benchmark, the observed effect of QLoRA under No RAG varied by model family and scenario. Answer Relevancy increased in all four model-scenario comparisons. Answer Correctness decreased for both model families in the obstetric challenge set and for MedGemma 1.5 in the MaternaQA-es subset; Semantic Similarity decreased for Gemma 4 in the challenge set but increased for MedGemma 1.5 and for both models in MaternaQA-es.

Within the evaluated questions, medical pretraining did not show a uniform advantage after domain-specific adaptation across the tested metrics and configurations. The highest Answer Relevancy under No RAG came from Gemma 4 QLoRA in MaternaQA-es and MedGemma 1.5 QLoRA in the obstetric challenge set. Gemma 4 Base with Hybrid retrieval achieved the highest Answer Correctness in both scenarios. The highest Semantic Similarity came from Gemma 4 QLoRA with Hybrid retrieval in MaternaQA-es and was tied between Gemma 4 Base and QLoRA with HyDE in the challenge set. These findings support evaluating multiple dimensions rather than treating a single automatic score as a complete measure of quality.

Hybrid retrieval increased Answer Correctness in all four model conditions in each scenario but reduced Answer Relevancy in six of eight comparisons. One plausible interpretation is that retrieved evidence increased agreement with reference answers while sometimes reducing direct focus. Retrieval-grounding scores were generally high for Context Precision but more variable for Context Recall, indicating that relevant evidence was often surfaced without consistently covering all reference-supporting information.

HyDE produced consistently high RAGAS Context Precision in both scenarios, but its effects on final answers were heterogeneous. In MaternaQA-es, it improved Context Precision relative to Hybrid in three of four model states and improved Answer Correctness over Hybrid only for Gemma 4 QLoRA. In the obstetric challenge set, it improved Context Precision for all four states, but increased Answer Correctness over Hybrid only for Gemma 4 QLoRA. Higher Context Precision therefore did not guarantee better final answers.

The auxiliary cases (Table V) combine supported core answers with unsupported additions or repetition; they establish neither overfitting nor a causal explanation for the retrieval differences.

V. Conclusions

The results suggest distinct roles for the three inference strategies rather than a single best configuration. Retrieval with Hybrid is the strongest starting point for agreement with the reference responses in this benchmark: it increased Answer Correctness relative to No RAG in all eight model-state-scenario comparisons, and Gemma 4 Base with Hybrid achieved the highest mean Answer Correctness in each scenario. This gain should be weighed against the lower Answer Relevancy observed in six of eight comparisons, which may indicate that additional evidence makes answers less direct. These results apply to a setting in which the relevant source documents are present in the retrieval corpus.

HyDE achieved higher Context Precision than Hybrid in seven of eight comparisons, but this advantage did not carry over to the final answers: it improved Answer Correctness over Hybrid only for Gemma 4 QLoRA, lowered Context Recall in the obstetric challenge set, and added a generation step (18.5 s on average per unique HyDE generation). In this benchmark, therefore, the results give no basis for preferring HyDE over Hybrid; it would warrant consideration only if evaluation on the target questions confirmed a gain in the final answers.

No RAG, the control for parametric knowledge, obtained the lowest Answer Correctness in six of eight comparisons and the highest Answer Relevancy in six of eight, including the two highest values overall. It remains relevant when an external corpus is unavailable or when response directness is the priority, but Answer Relevancy alone is not a sufficient basis for preferring it, because retrieval generally improved agreement with the reference responses.

The comparison also gives no basis for assuming that medical pretraining or QLoRA will lead across every objective. QLoRA increased No RAG Answer Relevancy in all four model-scenario comparisons, while its effects on Answer Correctness and Semantic Similarity varied. In practice, model adaptation and retrieval strategy should be selected according to the system's priority: reference alignment, response directness, or retrieval precision. These findings are descriptive results from 10 questions per scenario and automated metrics; they

TABLE IV
ANSWER CORRECTNESS: MEAN AND APPROXIMATE MARGINAL 95% CONFIDENCE INTERVAL

Model and state	No RAG	Hybrid	HyDE
MaternaQA-es test subset			
Gemma 4 Base	0.372 [0.238, 0.506]	0.649 [0.456, 0.841]	0.560 [0.375, 0.744]
Gemma 4 QLoRA	0.384 [0.269, 0.500]	0.605 [0.422, 0.788]	0.637 [0.441, 0.833]
MedGemma 1.5 Base	0.388 [0.258, 0.517]	0.599 [0.398, 0.801]	0.595 [0.387, 0.803]
MedGemma 1.5 QLoRA	0.330 [0.213, 0.446]	0.609 [0.454, 0.764]	0.574 [0.401, 0.747]
Obstetric challenge set			
Gemma 4 Base	0.359 [0.159, 0.558]	0.457 [0.244, 0.671]	0.335 [0.125, 0.545]
Gemma 4 QLoRA	0.205 [0.127, 0.282]	0.349 [0.185, 0.513]	0.399 [0.254, 0.544]
MedGemma 1.5 Base	0.287 [0.164, 0.411]	0.402 [0.139, 0.664]	0.260 [0.123, 0.396]
MedGemma 1.5 QLoRA	0.234 [0.152, 0.315]	0.455 [0.249, 0.660]	0.316 [0.098, 0.533]

Note. Mean [95% CI]; $n = 10$. Marginal t intervals assume independent questions; they do not test paired differences.

TABLE V
ILLUSTRATIVE AUXILIARY AUDIT OF ARCHIVED SOURCE-GROUNDED RESPONSES

Case and question focus	Supplied evidence / base response	QLoRA response	Observed issue
E1: CAMVA-COVID-19 study objective (Gemma 4)	The excerpt and base answer describe assessing menstrual changes and dysmenorrhea.	Adds an unsupported phase-II randomized trial with 1,200 women and 12 fertility centers.	Unsupported study-design details after adaptation.
E2: Timing of folic-acid initiation (MedGemma 1.5)	The source specifies early/preconception initiation; Base adds 4 mg/day, absent from that excerpt.	Keeps the supported timing without the added dose, but repeats the recommendation.	Better source alignment with repetitive output.
E3: Rationale for more prenatal visits (Gemma 4)	The source and Base cite perinatal-mortality concerns with fewer visits and psychosocial care.	Adds delivery-site continuity and a claim of no maternal/neonatal outcome differences.	Unsupported additions and conflicting outcome statements.

Note. Purposive auxiliary examples do not estimate error rates. Current Hybrid/HyDE responses are unavailable; these cases do not establish retrieval-specific errors.

can guide configuration choices for further evaluation, but do not establish clinical reliability. A broader question set and clinician assessment are needed before considering clinical use.

VI. LIMITATIONS AND FUTURE WORK

The evaluation covers 10 questions per scenario. Its approximate marginal intervals do not resolve paired statistical significance or account for document clustering, subset-selection bias, or evaluator variability. The auxiliary grounded audit cannot replace matched Hybrid/HyDE error analysis without current question-level run files. Automatic RAGAS scores require complementary clinician assessment and broader question coverage.

REFERENCES

[1] B. Pingua, A. Sahoo, M. Kandpal, D. Murmu, J. Rautaray, R. K. Barik, and M. J. Saikia, "Medical LLMs: Fine-tuning vs. retrieval-augmented generation," *Bioengineering*, vol. 12, no. 7, p. 687, 2025, doi: 10.3390/bioengineering12070687.

[2] Gemma Team, T. Mesnard, C. Hardin *et al.*, "Gemma: Open models based on gemini research and technology," arXiv:2403.08295, 2024.

[3] A. Bora and H. Cuayáhuitl, "Systematic analysis of retrieval-augmented generation-based LLMs for medical chatbot applications," *Machine Learning and Knowledge Extraction*, vol. 6, no. 4, pp. 2355–2374, 2024, doi: 10.3390/make6040116.

[4] K. Arzideh *et al.*, "Improving retrieval augmented generation for health care by fine-tuning clinical embedding models: Development and evaluation study," *Journal of Medical Internet Research*, vol. 28, p. e82997, 2026, doi: 10.2196/82997.

[5] I. Psilopatis, K. Redling, V. Filippi *et al.*, "The role of large language models in the diagnosis and management of obstetric patients: A pilot feasibility study using simulated obstetric cases," *Archives of Gynecology and Obstetrics*, vol. 313, p. 174, 2026, doi: 10.1007/s00404-026-08423-1.

[6] H. Ç. Arslan, K. Arslan, M. Gün *et al.*, "Evaluation of ChatGPT-5 responses in obstetric and gynecological emergencies: Concordance, readability, and clinical reliability," *BMC Emergency Medicine*, vol. 25, p. 220, 2025, doi: 10.1186/s12873-025-01396-5.

[7] G.-X. Xu, H. Jiang, S. V. Nguyen, D. Wu, Z.-D. Liu, F. Sha, and Y. Li, "Obstet-LLM: Large language model for early prediction of SGA-LGA newborns," *Journal of Computer Science and Technology*, vol. 41, no. 2, pp. 638–652, 2026, doi: 10.1007/s11390-025-5020-0.

[8] A. Noguera, A. L. Mogollón-Benavides, M. D. Niño-Mojica, S. Rúa, D. Sanín-Villa, and J. C. Tejada, "Applications and challenges of retrieval-augmented generation (RAG) in maternal health: A multi-axial review of the state of the art in biomedical QA with LLMs," *Sci*, vol. 7, no. 4, p. 148, 2025, doi: 10.3390/sci7040148.

[9] J. Pacheco-Romero, Í. Sandoval Díaz, and J. Ramos-Castillo, "Inteligencia artificial en la práctica de la ginecobstetricia, la investigación y redacción científica," *Revista Peruana de Ginecología y Obstetricia*, vol. 69, no. 4, 2023, doi: 10.31403/rpgo.v69i2566.

[10] Institución Universitaria de Envigado. (2026) MaternaQA-es. Data set. Accessed: Jul. 22, 2026. [Online]. Available: https://huggingface.co/datasets/iue-edu/MaternaQA-es

[11] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," arXiv:2305.14314, 2023.

[12] P. Lewis, E. Perez, A. Piktus *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.

[13] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "BGE M3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," arXiv:2402.03216, 2024.

[14] LanceDB. (2026) Vector search. LanceDB documentation. Accessed: Sep. 24, 2026. [Online]. Available: https://docs.lancedb.com/search/vector-search

[15] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proc. 32nd Int. ACM SIGIR Conf. Res. Develop. Inf. Retrieval*, 2009, pp. 758–759, doi: 10.1145/1571941.1572114.

[16] L. Gao, X. Ma, J. Lin, and J. Callan, "Precise zero-shot dense retrieval without relevance labels," in *Proc. 61st Annu. Meeting Assoc. Comput. Linguistics (Volume 1: Long Papers)*, 2023, pp. 1762–1777, doi: 10.18653/v1/2023.acl-long.99.

[17] Z. Ji, N. Lee, R. M. Frieske *et al.*, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, p. Article 248, 2023, doi: 10.1145/3571730.

[18] S. Es, J. James, L. Espinosa Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," in *Proc. 18th*

Conf. Eur. Chapter Assoc. Comput. Linguistics: System Demonstrations, 2024, pp. 150–158, doi: 10.18653/v1/2024.eacl-demo.16.

[19] J. Gallifant, M. Afshar, S. Ameen *et al.*, "The TRIPOD-LLM reporting guideline for studies using large language models," *Nature Medicine*, vol. 31, no. 1, pp. 60–69, 2025, doi: 10.1038/s41591-024-03425-5.

[20] E. M. Bender and B. Friedman, "Data statements for natural language processing: Toward mitigating system bias and enabling better science," *Transactions of the Association for Computational Linguistics*, vol. 6, pp. 587–604, 2018, doi: 10.1162/tacl_a_00041.